# Parul Singh

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## EDUCATION

BTech (CSE)- Chhatrapati Shahu Ji Maharaj University, Kanpur (CPI: 8.62/10) 2021-2025

## EXPERIENCE

**Indian Institute of Technology, Kanpur** IITK Campus, Kanpur
Full Stack Web Developer, Research Intern, May 2024 - July 2024

- Designed and implemented a Django application that seamlessly integrated with six regulatory bodies, eliminating the need for manual document searches and saving the compliance team an estimated 20 hours per week on documentation tasks.
- Developed an automated system for fetching and updating regulatory documents, which is now utilized by a team; this initiative improved efficiency and minimized organizational risks.

## PROJECTS

**Automation of Regulatory updates for the Indian Power Sector** **GitHub Link**
Prof. Anoop Singh/Surge Program 2024 May 2024 - July 2024

- Developed a comprehensive regulatory documents tracker tailored for the Indian Power Sector. This tool streamlines the process of monitoring and managing regulatory documents, ensuring efficient compliance and easy access to critical information**.**
- This solution can be integrated with multiple regulatory bodies, eliminating the need for manual document searches and significantly reducing time expenditure.
- **Tech Stack:** HTML, CSS, JavaScript, Python libraries (beautifulSoup, requests, smtplib), Django, MySQL, Web Scraping

**Book Recommender System** **GitHub Link**
Self-Made Sept 2025

- Built a machine learning-based recommendation system that suggests books to users using collaborative filtering and popularity-based approaches.
- Enhanced personalization and improved user experience by analyzing user preferences and large datasets.
- **Tech Stack:** Python, Pandas, NumPy, Scikit-learn, Streamlit

**Solar Panel Defect Classification System** **GitHub Link**
Self-Made Mar 2026

- Developed an AI-powered web application to detect and classify solar panel defects using a fine-tuned EfficientNet model.
- Built an interactive Streamlit interface that allows users to upload images and receive real-time predictions with confidence scores and probability distributions.
- Implemented image preprocessing and deep learning inference pipeline, achieving accurate multi-class classification across 6 defect categories.
- Enhanced user experience with visual insights, including top predictions and class probability graphs.
- **Tech Stack:** Python, TensorFlow/Keras, Streamlit, NumPy, Pandas, Plotly

## TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, SQL, HTML5, CSS3
- **Frameworks & Libraries:** Django, Bootstrap, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, NLTK, TensorFlow, Keras
- **Machine Learning & AI:** Machine Learning, Supervised & Unsupervised Learning, Natural Language Processing (NLP), Deep Learning, Convolutional Neural Networks (CNN), Model Evaluation, Feature Engineering
- **Data Analysis & Visualization:** Pandas, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Data Cleaning.
- **Web & Backend Development:** Web Scraping (BeautifulSoup, Selenium, Requests)
- **Tools & Version Control**: Git, GitHub, Streamlit

## POSITION OF RESPONSIBILITY

- Worked in a team of two during the IITK internship. May 2024 - July 2024

## ACHIEVEMENTS

- Selected under the SURGE Program 2024 and nominated for the Institute Funded Category for a stipend.
- Selected in Round 1 of Unstop Playstorm for presenting “The Glitch Room” – a Gen Z-focused engagement idea using gamification, storytelling, and chaos-led innovation.
- Participated in the Google Cloud Arcade Facilitator Program 2025 and reached the **Arcade Novice Tier.**